

From Face Perception to Social Categorization: A Multidimensional Scaling Approach to Ethnic Outgroup Homogeneity in Childhood

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The Outgroup Homogeneity Effect

Outgroup members appear more homogeneous to each other than ingroup members

- Supported in adult populations

The Outgroup Homogeneity Effect

Early Emergence of Categorization vs. Childhood OHE Gap

- **In infancy:** Strong evidence for ingroup preferences and visual specialization for ingroup faces (Kelly et al., 2009; Quinn et al., 2016).
- **In childhood:** No consolidated evidence for the Outgroup Homogeneity Effect (OHE). Why?

A few structural obstacles

1. Using a explicit tasks it is difficult not to create top-down categories
2. Most of the literature employs a dyadic paradigm (ingroup-outgroup)
3. If visual paradigms are used, the stimuli selection is subjected to biases

1. Spontaneous bottom-up categorization

Multidimensional Scaling (MDS)

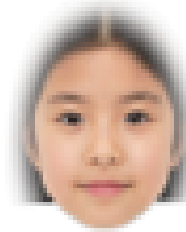
- Face-Space Theory (Valentine et al., 2016): Faces exist in a continuous multidimensional perceptual space
- **Goal:** Visually map whether faces from different ethnic groups are spontaneously perceived as more similar to one another (OHE), and how this changes across age groups.

2. Multi-group context

- 4-group context: Italian majority-ingroup + 3 distinct ethnic minority-outgroups (African Descent, East Asian, Andean)
- 20 faces per gender group (5 per ethnic group)

3. The AI generated-stimuli

- Morphologies controlled using an emotional filter and FaceNet embeddings to ensure perceptual distinctiveness

African Descent	East-Asian	Italian	Andean
			

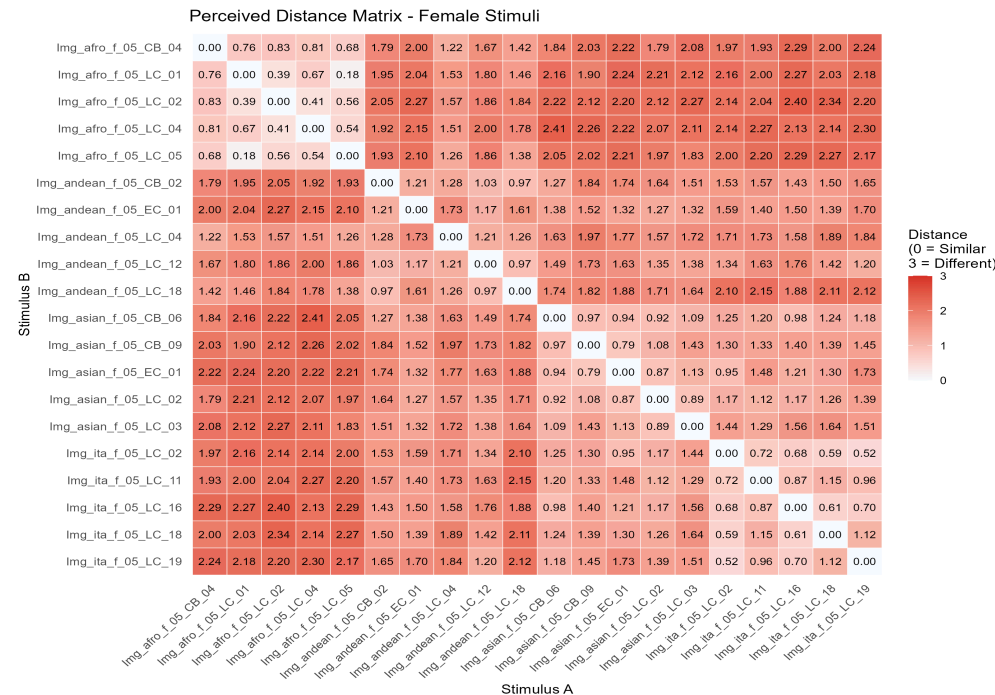
The Face Similarity Task

→	→	→
		Quanto si somigliano? Molto Diversi Un po' Diversi Un po' Simili Molto Simili
800ms	1000ms	//

The Face Similarity Task

- Incomplete block design
- 66 random dyads presented sequentially (3 faces per group)

Moderated (S1)



Pre-registered studies: Method & Experimental Design

Sample:

- Study 1 (moderated, controlled school setting): $N=249$, $M=5.17$ years, $SD=1.23$, 134 boys (53.8%) and 115 girls (46.2%)
- Study 2 (unmoderated, online setting): $N=101$, $M = 6.06$, $SD=1.43$, 52 girls (51.5%) and 49 boys (48.5%)
- Age range: 3 to 8 years

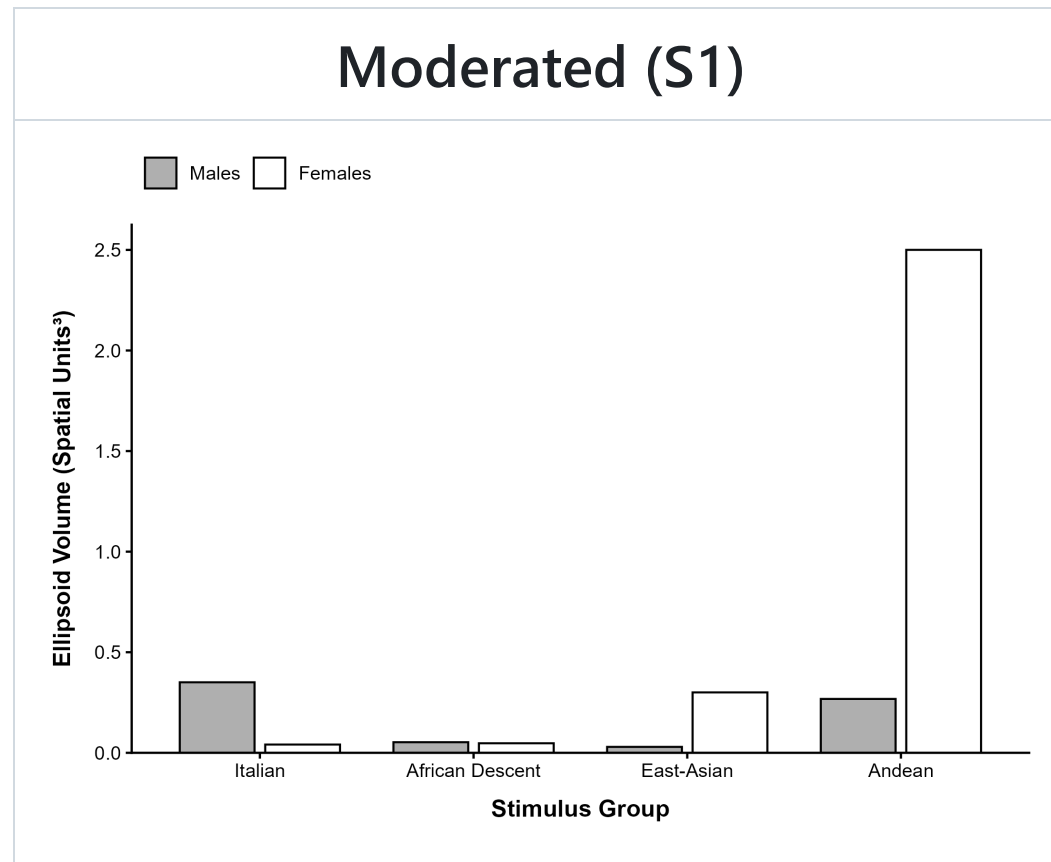
MDS

3D MDS visualization

- Children spontaneously visually categorize stimuli in ethnic groups

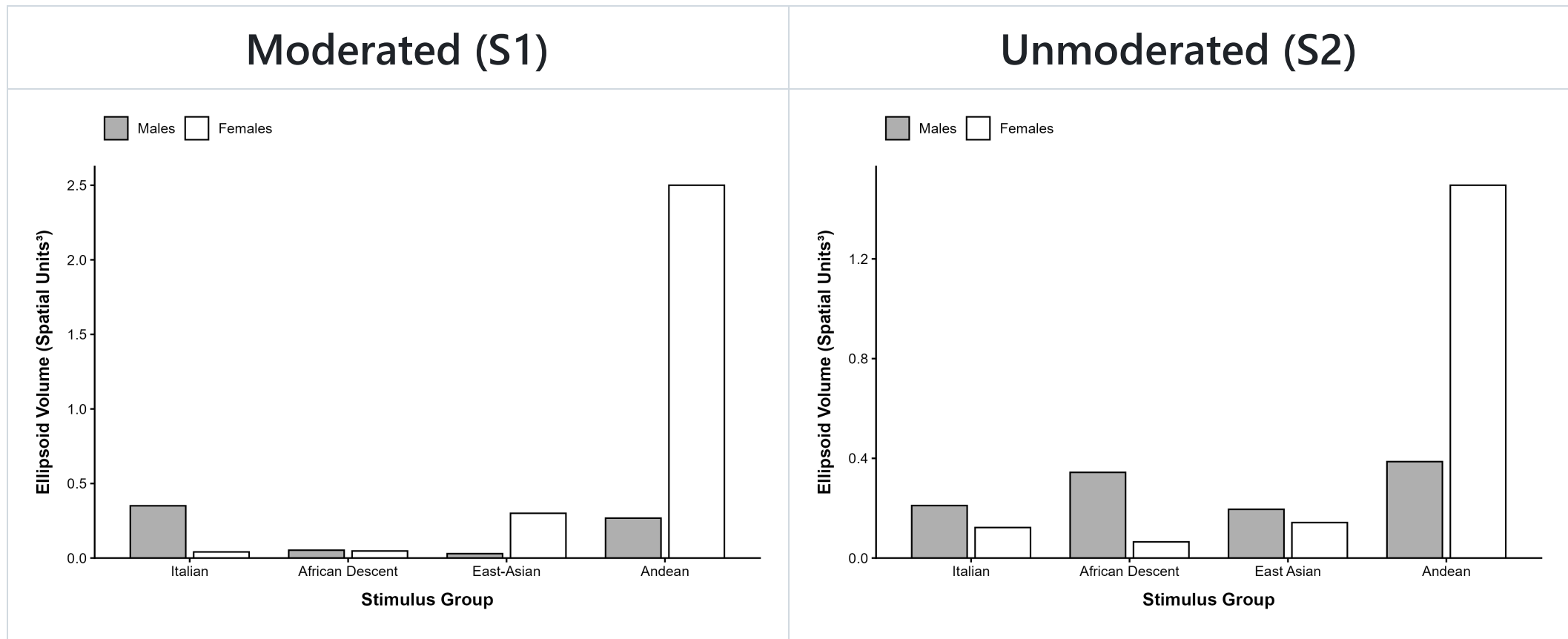
Outgroup Homogeneity Effect

- Ellipsoid's volume:



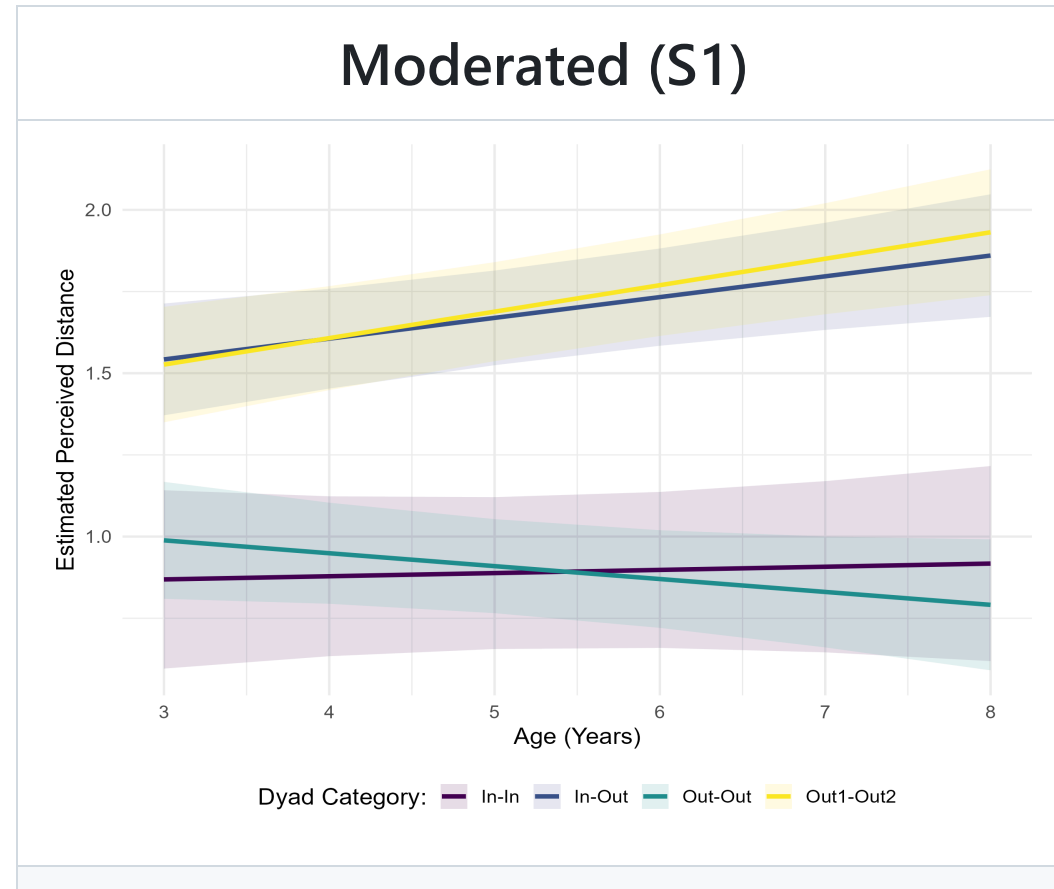
Outgroup Homogeneity Effect

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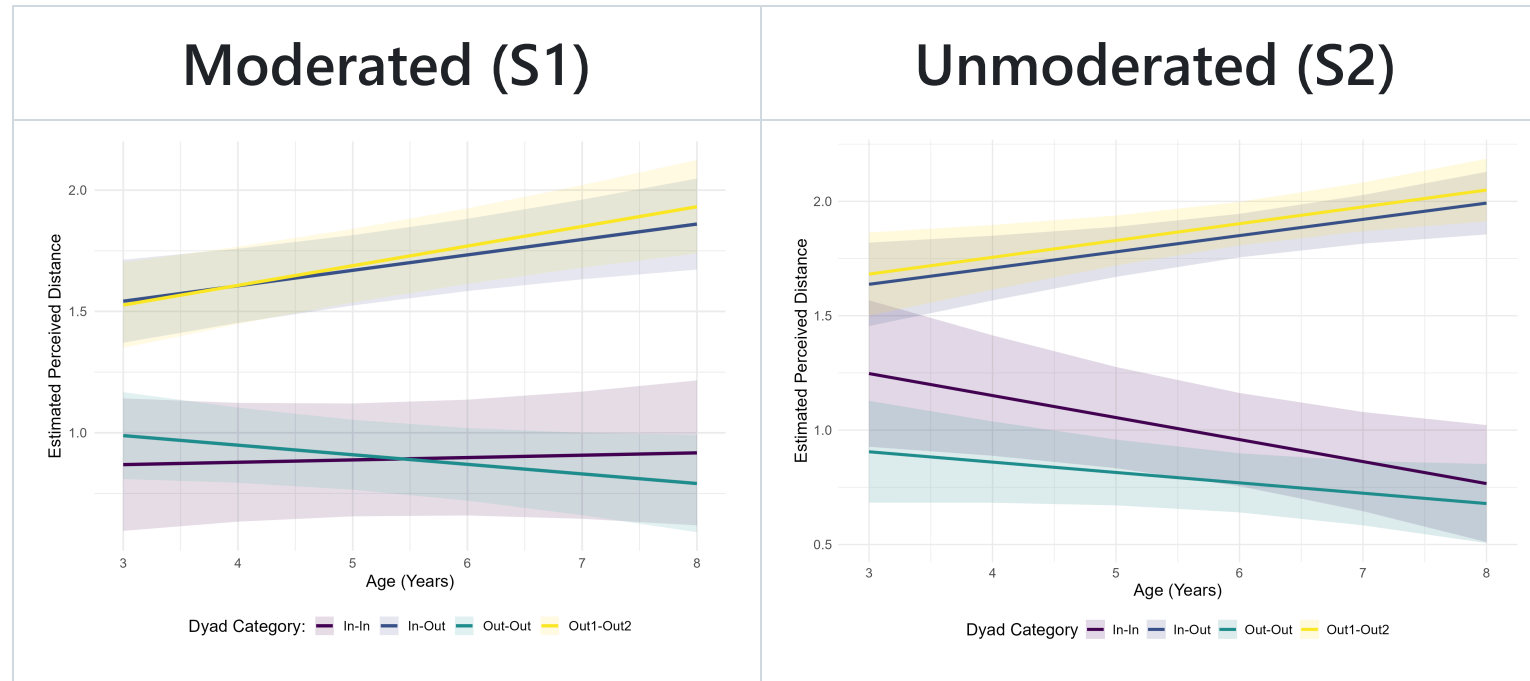
LMM - Developmental Trajectories

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- In-Out --> $b = 0.06$, $SE = 0.02$, 95% CI [0.02, 0.11]
- Out1-Out2--> $b = 0.08$, $SE = 0.02$, 95% CI [0.04, 0.12]

LMM - Developmental Trajectories



Dyad type	<i>b</i>	SE	95% CI lower	95% CI upper
In-Out	0.06	0.02	0.02	0.11
Out1-Out2	0.08	0.02	0.04	0.12
In-In	0.01	0.03	-0.06	0.08
In-In	-0.04	0.03	-0.09	0.01

- Intergroup similarity judgments increased with age; within-ingroup judgments did not show reliable change.

Conclusion

- Children spontaneously visually categorize stimuli in ethnic groups
- Intergroup similarity judgments increased with age; within-ingroup judgments did not show reliable change.

Questions? 🙌

PRESENTATION AVAILABLE AT

